

A Case-Study on the Effect of Wearing Sunglasses on the Classification Accuracy of SSVEP-BMI

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SSVEP (Steady-State Visual Evoked Potential) is a type of visual evoked potential that occurs in the visual cortex in response to periodic visual stimulation, exhibiting frequency characteristics that match the stimulus frequency. However, SSVEP has certain drawbacks, including increased eye strain and discomfort due to prolonged exposure to flickering visual stimuli, as well as susceptibility to external light interference. To address these issues, we explored the feasibility of measuring SSVEP while wearing sunglasses. In this study, we compared the classification accuracy of machine learning models under conditions with and without sunglasses to evaluate their impact on SSVEP-based classification.

Key Words : SSVEP, BMI, sunglasses

1. Introduction

As of 2025, advanced countries worldwide are experiencing an accelerating aging population, a trend projected to continue. Japan's total population was approximately 124.35 million as of October 1, 2023, with 36.23 million people aged 65 or older, constituting 29.1% of the total population. Concurrently, Japan faces societal challenges such as a declining birthrate. This combination is expected to lead to a shortage of caregivers, consequently diminishing the Quality of Life (QoL) for individuals requiring nursing care, particularly the elderly with physical disabilities. To mitigate this issue, research into Brain-Computer Interfaces (BCI) and Brain-Machine Interfaces (BMI), which enable the operation of devices and computers solely through brainwaves without physical movement, is gaining significant attention.

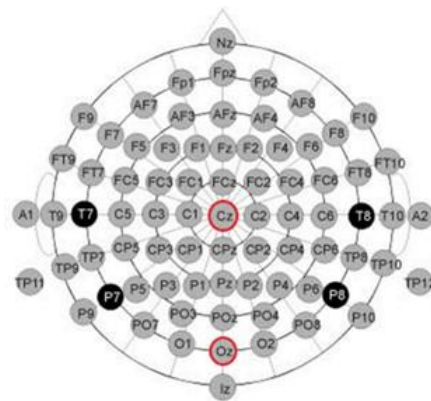
Brainwaves encompass various types associated with different phenomena, including SSVEP (Steady-State Visual Evoked Potential), MI (Motor Imagery), and P300. Among these, this study focuses on SSVEP. SSVEP is a visual evoked potential generated in the visual cortex, and other brain regions, in response to visual stimulation flickering at a specific frequency. It manifests as a steady sinusoidal potential with the same frequency as the stimulus, as well as its integer multiples. For instance, if a 15 Hz visual stimulus is presented, brainwave components at 15 Hz will be amplified synchronously. The advantages of SSVEP include ease of feature extraction, short trial times allowing for minimal delay, and a reduced need for habituation or training. Conversely, disadvantages of SSVEP include its dependence on ambient illumination and the high visual load it imposes.

This research specifically addresses the two drawbacks of SSVEP: its dependence on ambient illumination and the high visual load.

Previous studies have suggested that SSVEP classification accuracy can decrease due to ambient light conditions. Furthermore, many studies have identified discomfort and fatigue from prolonged visual stimulation as significant challenges. As a potential solution to these problems, presenting visual stimuli while wearing sunglasses has been proposed. Wearing sunglasses is expected to help maintain consistent ambient illumination and reduce discomfort and fatigue associated with visual stimulation. Therefore, the objective of this study is to compare SSVEP classification accuracy when wearing sunglasses versus not wearing them.

2. Experimental Conditions

Unicorn Hybrid Black manufactured by Guger Technology, with g.SAHARA Active electrodes from the same company. Five subjects (aged 21-22) participated in the study; none had known neurological disorders or significant health issues. The electrodes were positioned as shown in Figure 1.



3. Method

The experimental procedure is outlined below:

- ① Subjects were seated in a comfortable posture in a chair.
- ② A black and white checkerboard pattern (Figure 2) was presented at the center of the display, and subjects were instructed to fixate on its central part.
- ③ After fixating on the image for 30 seconds, a 30-second interval was introduced.
- ④ Step 3 constituted one trial, and one set consisted of 18 trials (6 trials for each frequency: 9, 12, 15 Hz).
- ⑤ Steps 1-4 were performed for three conditions: display only (no sunglasses), light sunglasses, and dark sunglasses. A total of 54 trials were conducted for each subject on a single day.
- ⑥ After the experiment, a questionnaire was administered, asking subjects to rate their discomfort level towards the visual stimuli on a scale of 1 to 10 across the three conditions.

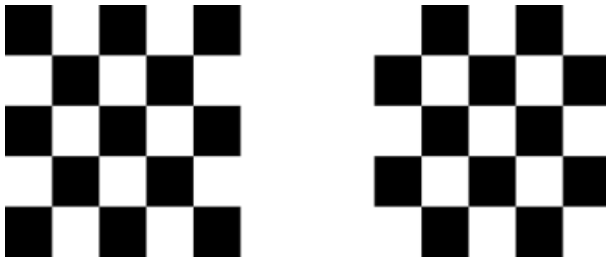


Fig 2. Black and white checkerboard

4. Results

(a) Classification Accuracy Results

The classification accuracy results obtained from machine learning on data measured from 5 subjects are shown below.

Table 1. Classification accuracy and average

	A	B	C	D	E	AVE
no sunglasses	98.11	97.79	98.11	97.79	98.42	98.04
light sunglasses	96.53	96.53	97.79	98.10	99.37	97.66
dark sunglasses	98.42	96.53	96.21	96.53	97.79	97.10

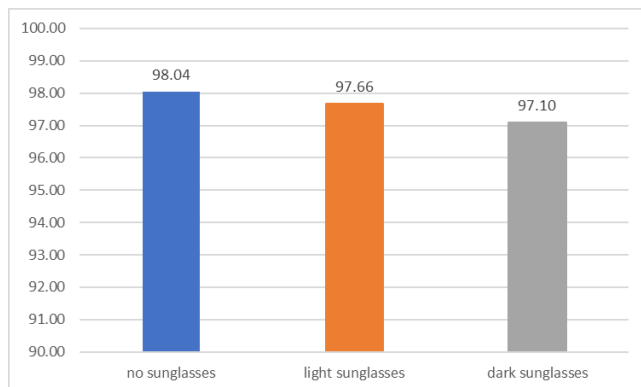


Fig 3. Classification average

(b) Questionnaire Results

The questionnaire results regarding discomfort after the experiment are presented below.

Table 2. Survey results and average

	A	B	C	D	E	AVE
no sunglasses	5	8	8	7	7	7
light sunglasses	3	6	6	4	6	5
dark sunglasses	1	3	4	1	4	2.6

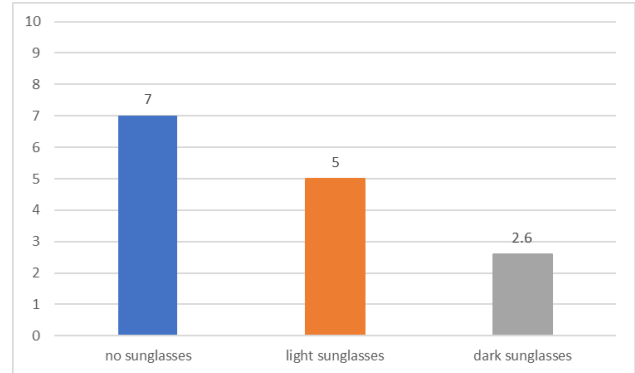


Fig 4. Survey results average

5. Discussions and Conclusions

We investigated the effect of wearing sunglasses on SSVEP classification accuracy. As a result, high classification accuracies of over 97% were obtained for all three conditions. The difference between the highest classification accuracy (display only) and the lowest (dark sunglasses) was approximately 1%. Even with different concentrations of sunglasses, the classification accuracy remained consistently high, with a difference of about 0.5%. This indicates that the classification accuracy when wearing sunglasses is not significantly different from that without sunglasses. Next, the questionnaire results from section 4.(2) revealed that SSVEP visual stimulation caused considerable discomfort, and wearing sunglasses was shown to have a discomfort-reducing effect. This suggests that wearing sunglasses is effective in alleviating eye strain and discomfort.

References

- [1] 内閣府「1 高齢化の現状と将来像」『内閣府 Cabinet Office』, URL: https://www8.cao.go.jp/kourei/whitepaper/w-2024/html/zenbun/s1_1_1.html, date 2025-01-13.
- [2] R. Zhang et al., "Improving AR-SSVEP Recognition Accuracy Under High Ambient Brightness Through Iterative Learning," in IEEE Transactions on Neural Systems and Rehabilitation Engineering, vol.31, pp. 1796-1806, 2023.
- [3] A. González-Mendoza, J. L. Pérez-Benítez, J. A. Pérez-Benítez and J. H. Espina- Hernández, "Brain Computer Interface based on SSVEP for controlling a remote control car," 2015 International Conference on Electronics, Communications and Computers (CONIELECOMP), pp. 93-97, 2015.
- [4] 阿部和穂「『ブロードマンの脳地図』の謎と魅力...脳科学者を魅了する理由」『All About 健康・医療』<https://allabout.co.jp/gm/gc/491827/>, date 2025-01-13